

Mary Chege
Ph.D. Candidate
One Health Centre,
Kenya Institute of Primate Research (KIPRE) Nairobi, Kenya

mchege89@gmail.com
mchege@nd.edu

EDUCATION

University of Nairobi (UoN) & University of Notre Dame **Nairobi, Kenya**
Ph.D., Bioinformatics May 2023 – December 2026 (Expected)
Supervisors:
Dr. Rosaline Macharia, Prof. George Obiero,
Dr. Mercy Akinyi, Prof. Elizabeth Archie

Jomo Kenyatta University of Agriculture and Technology (JKUAT) **Juja, Kenya**
MSc, Bioinformatics and Molecular Biology Nov 2018
Supervisors: Prof. Evans Nyaboga, Prof. Fred Wamunyokoli, Dr. Joseph Kamau

Jomo Kenyatta University of Agriculture and Technology (JKUAT) **Juja, Kenya**
BSc, Biochemistry and Molecular Biology June 2013
Supervisor: Dr. Joseph Ng'ang'a

EMPLOYMENT

Amboseli Baboon Research Project (ABRP) **Nairobi, Kenya**
Ph.D. Research Fellow May 2023 - Present
Advisor: Prof. Elizabeth Archie

Kenya Institute of Primate Research (KIPRE) **Nairobi, Kenya**
Postgraduate Research Fellow Aug 2021 – present
Advisors: Dr. Joseph Kamau

Jomo Kenyatta University of Agriculture and Technology **Juja, Kenya**
Part-time Lecturer Aug 2016 – Dec 2019

Pwani University **Kilifi, Kenya**
Part-time Lecturer Aug 2016 – Dec 2019

RESEARCH INTERESTS

I use metagenomics and molecular biology approaches to study:

- The interplay between social grouping, seasonal variations, and gut microbiome composition in wild baboons and livestock.
- Antimicrobial resistance (AMR) gene transmission and pathogen surveillance at the human-wildlife-environment interface.

RESEARCH EXPERIENCE

Amboseli Baboon Research Project (ABRP) **2023 – Present**
Ph.D. Research Fellow | Amboseli Ecosystem, Kenya
For my doctoral research, I investigate the eco-evolutionary dynamics of the gut microbiome in wild baboons and livestock using a One Health approach. My work spans from field collection and sequencing preparation to advanced computational analysis. On high-performance computing (HPC) clusters, I process large-scale metagenomic datasets and construct statistical pipelines to integrate microbial community profiles with multi-decade behavioral matrices provided by our international ABRP collaborators.

POSTGRADUATE & INSTITUTIONAL EXPERIENCE

Kenya Institute of Primate Research (KIPRE) — One Health Centre

Postgraduate Research Fellow | Nairobi, Kenya

2021 – Present

Assist in the coordination of research workflows across multiple active molecular diagnostic and validation initiatives. Responsible for standardizing field sample collection grids, ensuring laboratory biosafety compliance, and managing centralized operational data pipelines to achieve project goals across multi-institutional collaborations.

University of Florida & Kenya Institute of Primate Research

Research Fellow (Tick-Borne Pathogen Screening Initiative) | Northern Kenya

July 2023 – October 2023

Under the joint direction of **Prof. Michael Von Fricken** and **Dr. Joseph Kamau**, I assisted in the execution of field-forward molecular surveillance networks tracking zoonotic tick-borne pathogens across remote field stations in Northern Kenya. Supervised on-site vector processing pipelines.

Smithsonian Conservation Biology Institute & KIPRE One Health Centre

Research Fellow (Bat-Borne Viral Pathogen Surveillance) | Western Kenya

2020 – 2021

Contributed to a federally funded U.S. government initiative directed by **Dr. Dawn Zimmerman** and **Dr. Joseph Kamau**. The program was established to perform deep-cave wildlife tracking and biosurveillance, specifically optimizing molecular assays to detect novel coronaviruses (including SARS-CoV-2) and broader viral cross-sections within wild bat populations across Western Kenya to surveil emerging and re-emerging infectious zoonotic disease risks.

George Mason University & KIPRE One Health Centre

Research and Technical Assistant (REDI-NET Program)

April 2022 – July 2022

Mpala Research Centre, Laikipia, Kenya

Operated as a primary field and laboratory technical analyst for the Remote Emerging Disease Intelligence Network (REDI-NET), a federally funded U.S. government consortium managed alongside **Prof. Michael Von Fricken** and **Dr. Joseph Kamau**. The initiative focused on standardizing field-forward pathogen detection workflows across environmental vectors at high-density human-wildlife-environmental interfaces. Installed and calibrated automated extraction platforms, optimized laboratory documentation sheets for cross-border reproducibility, and systematically audited infrastructure gaps in remote settings.

KIPRE One Health Centre

Laboratory & Data Analyst (COVID-19 Detection & Surveillance) | Nairobi, Kenya

2020 – 2022

Served as a core molecular diagnostics analyst within the high-volume national pandemic surveillance pipeline. Managed secure clinical sample networks, verified chain-of-custody diagnostics records, and maintained operational high-throughput screening data systems.

PUBLICATIONS

Publications in peer-reviewed journals:

- 2026 Southworth CA, Winans JC, Gordon JB, Learn NH, Wilber WA, Andreadis C, Andreassen G, Arandjelovic M, Campbell CR, Chege MN, Creighton MJA, Cromer CM, Debray R, Dickson CC, Ferretti P, George EM, Gesquiere LR, He S, Hey L, Jefferson EE, Kulahci IG, Lerch BA, Nonnamaker L, Rivas-González I, Tuliozi B, Webb SE, Alberts SC, Archie EA, and Tung J. Demographic, behavioral, and ecological data from a long-term field study of wild baboons in Amboseli, Kenya. [Scientific Data](#)
- 2025 **Chege M**, Ferretti P, Webb S, Macharia RW, Obiero G, Kamau J, Alberts S, Tung J, Akinyi M, and Archie E. *Eukaryotic composition across seasons and social groups in the gut microbiota of wild baboons.* [Animal Microbiome](#)
- 2024 Obanda V, Akinyi M, King'ori E, Nyakundi R, Ochola G, Oreng P, Mugambi K, Waiguchu GM, Chege MN, Rosenbaum W, Ylitalo EB, Bäck AT, Pettersson L, Mukunzi OS, Agwanda B, Stenberg-Lewerin S, and Lwande OW. Epidemiology and ecology of the sylvatic cycle of African Swine Fever Virus in Kenya. [Virus Research](#)

- 2022 Kamau J, Ergunay K, Webala PW, Justi SA, Bourke BP, Kamau MW, Hassell J, **Chege MN**, Mwaura DK, Simiyu C, Kibiwot S, Onyuok S, Caicedo-Quiroga L, Li T, Zimmerman DM, and Linton Y-M. A Novel Coronavirus and a Broad Range of Viruses in Kenyan Cave Bats. *Viruses*
- 2017 **Chege MN**, Wamunyokoli F, Kamau J, and Nyaboga EN. Phenotypic and genotypic diversity of *Xanthomonas axonopodis* pv. *manihotis* causing bacterial blight disease of cassava in Kenya. *Journal of Applied Biology & Biotechnology*

FUNDING

Awarded:

Kenya Institute of Primate Research (KIPRE), Kenya.

2025-2026

PI: M. Chege

Characterizing gut microbiome and antimicrobial resistance in captive Baboons at the Kenya Institute of Primate Research

Total Award Amount: \$6,900

Max Planck Society – ARTEMIS Program

2025

Mentorship and scientific exchange with the Max Planck Institutes

Total Award Amount: 5,000€

Amboseli Baboon Research Project – Full Doctoral Fellowship

2023 – 2026

Full funding for PhD studies in Bioinformatics at the University of Nairobi

Total Award Amount: KES 1,324,000

PRESENTATIONS

Invited Talks

- Helmholtz Institute for One Health (HIOH), Greifswald, Germany, 2025
- ABRP Workshop, Max Planck Institute of Evolutionary Anthropology, Leipzig, Germany, 2025.
- Kenya Institute of Primate Research (KIPRE) Academic Seminar Series, Karen, Nairobi, Kenya, 2023.

Oral & Workshop Presentations (presenting author)

- Amboseli Baboon Research Project (ABRP) Workshop, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany, 2025.
- Amboseli Baboon Research Project (ABRP) Annual Workshop, University of Notre Dame, Notre Dame, Indiana, USA, 2023.
- Amboseli Baboon Research Project (ABRP) Joint Workshop, University of Nairobi (UoN), Nairobi, Kenya, 2023.

Posters (presenting author)

- Kenya Health Security Convention, Mombasa, Kenya, 2026.
- Microbiome Conference, Cold Spring Harbor Laboratory, NY, USA, 2024.
- Amboseli Baboon Research Project Workshop, University of Notre Dame, USA, 2023.
- Fourth International Cassava Conference, Cotonou, Benin, 2018.

Conference Attendance & Participation

- **Participant**, International Society for Microbial Ecology (ISME) Africa Symposium and Hands-on Technical Workshop, University Mohammed VI Polytechnic (UM6P), Benguerir, Morocco, 2025.
- **Participant**, 2nd International Wildlife Scientific Conference, Wildlife Research and Training Institute (WRTI), Naivasha, Kenya, 2025.

HONORS & AWARDS

ISME Africa Symposium Full Financial Support Award, Morocco, Dec 2025.

- ARTEMIS Mentorship Programme Award, Germany, 2024.
- NGS Academy/Africa CDC - Africa Pathogen Genomics Initiative Travel Grant, 2024.
- African Union / Africa CDC Travel Grant Award, 2023.

TECHNICAL EXPERTISE

Field & Laboratory Frameworks

Field Epidemiological Biosurveillance: Deep-cave mist-netting, handling, and non-invasive pathogen sampling in wild bat populations; non-invasive fecal sample collection from wild/captive baboons and sympatric livestock; multi-matrix environmental surface water sampling; tick vector field collection and identification.

Molecular Biology Wet Lab: High-yield manual and automated total nucleic acid (DNA/RNA) extraction tailored for diverse biological samples; conventional and quantitative real-time PCR (qPCR) assay setup and diagnostic runs.

Field Data Logistics: Remote deployment of digital mobile data collection forms via KoBoToolBox for real-time sample metadata logging; management of sample inventories using Microsoft Excel.

Bioinformatics & Computational Data Analytics

Computing & Infrastructure: Proficient in High-Performance Computing (HPC) cluster environments (SLURM/SGE schedulers), remote server management and synchronization via SSH, and architectural maintenance of multi-terabyte metagenomics datasets.

Operating Systems & Environment Control: Fluent execution within Unix/Linux CLI environments using custom Bash shell scripting; dependency control and package isolation via containerized Conda environments.

Workflow Automation: End-to-end pipeline automation, debugging, and utilizing Snakemake workflow.

Programming & Statistical Computing: High-dimensional scientific data parsing, quantitative statistical modeling, and publication-quality data visualization leveraging R and Python.

Data analysis: metagenomic data preprocessing, taxonomic and functional profiling; phylogenetic analyses and phylogenetic tree visualization

WORKSHOPS & TRAINING

- Certified Professional in Biorisk Management, International Federation of Biosafety Associations (IFBA), Professional Certification, 2024.
- Biorisk Risk Assessor Training, Foundational and Core Assessment Methodologies, IFBA Training Division, 2024.
- Trainers Development Program, Advanced pedagogical training and curriculum design for institutional biological safety deployment, IFBA, 2024.
- Next-Generation Sequencing (NGS) for Pathogen Surveillance Implementation Workshop, South African National Bioinformatics Institute (SANBI), University of the Western Cape, Cape Town, South Africa, 2024.
- Specialized Pathogen Genomic Surveillance Bioinformatics Training Workshop, South African National Bioinformatics Institute (SANBI), University of the Western Cape, Cape Town, South Africa, 2023

TEACHING EXPERIENCE

- **Tutor/Trainer**, Bioinformatics and Metagenomics, KIPRE. Organized and delivered training sessions including the Interuniversity Program Molecular Biology (IPMB), 2021 – Present.
- **Part-time Lecturer**, Biochemistry and Bioinformatics units, JKUAT & Pwani University, 2016 – 2019.

MENTORSHIP EXPERIENCE

Research Supervisor & Technical Mentor:

One Health Centre, Kenya Institute of Primate Research (KIPRE)

2021 – Present

- Directly supervise and provide practical mentorship to regional and visiting postgraduate trainees. Guide junior researchers through laboratory protocols, tracking their technical competency across molecular diagnostics (nucleic acid extraction, qPCR) and advanced bioinformatics pipelines for computational genomic analysis.

EXTERNAL SCIENTIFIC OUTREACH & SERVICE

Media & Interviews

- VICE News Documentary, featured for expertise in field-based pathogen detection in "[Hunting the Next Global Virus](#)."

Institutional Leadership

- **Class Representative**, JKUAT Union of Biochemists (JUBSC), 2009 – 2010. Represented student academic interests before departmental staff and committees.

Professional Memberships

- International Society for Microbial Ecology (ISME) Dec 2025 – Present
- Bioinformatics Hub of Kenya (BHKi) Jan 2020 – Present
- Organization of Women in Science in Developing Countries (OSWD). Nov 2019 – Present

Peer Review

- Invited reviewer for *Scientific Reports*. April 2026 – Present

REFERENCES

Dr. Joseph Kamau

Head of One Health Centre, Kenya Institute of Primate Research, Karen, Kenya

Phone: +254 723 432 901 jkamau@primateresearch.org

Prof. Elizabeth Archie

Department of Biological Sciences, University of Notre Dame, Notre Dame, IN, 46556, USA

Phone: +1 (574) 631-0178 - earchie@nd.edu

Dr. Mercy Akinyi

Ag. Director, Animal Sciences, Welfare and Ethics Directorate, Kenya Institute of Primate Research, Kenya

Phone: +254 722 394 428 - makinyi@primateresearch.org